

Produção de órgãos artificiais (3D)

Salford & Co.



The Science of Using Life to Improve Life

Biotechnology involves the use of living systems and organisms to develop or make useful products. From ancient fermentation to modern genetic engineering, it has evolved into a broad, multidisciplinary field.



What is Biotechnology?

Introduction

The Evolution of Biotechnology



Ancient Biotechnology

Use of yeast in bread and fermentation in alcohol.

19th Century

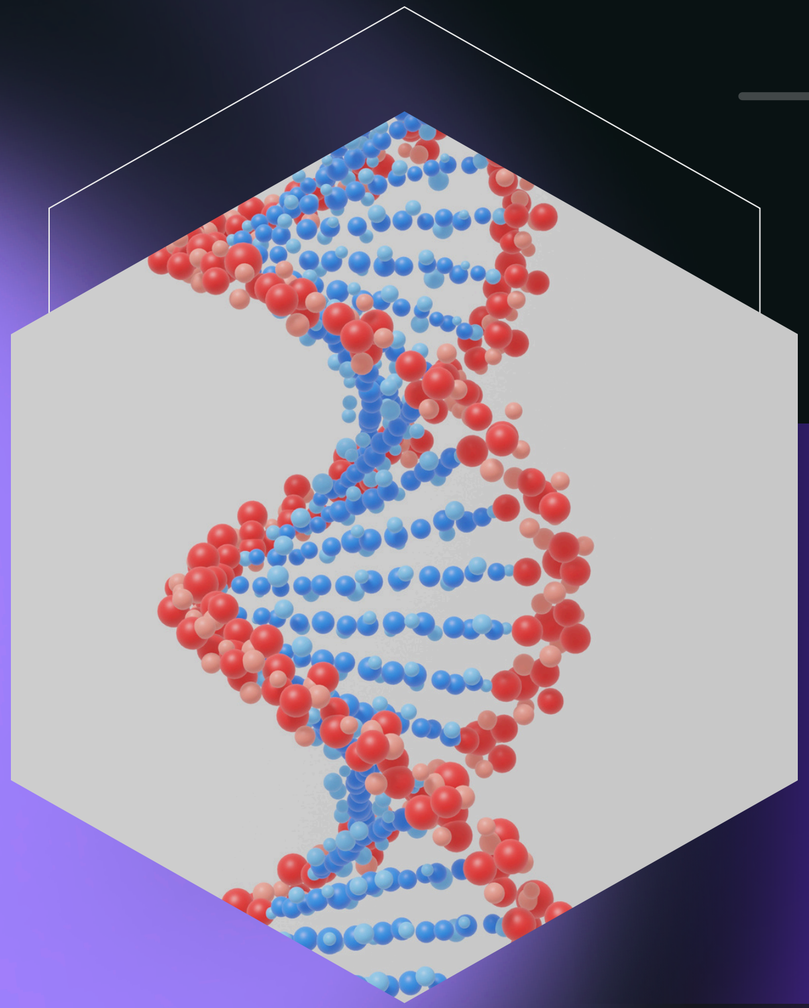
Discovery of microorganisms and basic genetics (Mendel's laws).

20th Century

DNA structure discovery, recombinant DNA, cloning.

21st Century

CRISPR, mRNA vaccines, personalized medicine, synthetic biology.



Core Fields in Biotechnology

Biotechnology is divided into several key sectors, each targeting specific areas of application. Understanding these branches helps clarify its broad influence.

- **Medical Biotechnology**
- **Agricultural Biotechnology**
- **Environmental Biotechnology**
- **Industrial Biotechnology**

Tools and Techniques

Genetic Engineering

Modifying DNA for desired traits in organisms.

Cell Culture

Growing cells in controlled environments for experimentation.

CRISPR-Cas9

A precise tool for gene editing and research.

Bioinformatics

Using software and algorithms to analyze genetic data.

PCR

Amplifies DNA for analysis.



Medical Applications

Transforming Healthcare Systems

Biotechnology has revolutionized medicine by enabling faster diagnostics, safer treatments, and even disease prevention.

Diagnostics

Regenerative Medicine

Personalized Medicine

Gene Therapy

Biopharmaceuticals



Industrial Biotechnology

Toward a Sustainable Industrial Future

Biofuels	Renewable energy from biomass, like ethanol and biodiesel.
Bioplastics	Eco-friendly plastics from natural materials.
Enzyme Engineering	Improving industrial processes with biological catalysts.
Bioreactors	Large-scale fermentation systems for production.



Current Innovations



CRISPR Advances

Enhanced precision and broader applications.



Synthetic Biology

Designing organisms for new functions.



3D Bioprinting

Printing tissues and organs using living cells.



Microbiome Research

Harnessing gut bacteria for health.

Escala Industrial e seus Desafios



- **Suprimento e expansão celular**
- **Vascularização e perfusão complexa**
- **Velocidade de bioimpressão e viabilidade celular**
- **Padronização e controle de qualidade**
- **Logística de Preservação**

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What’s Next in Biotechnology?



The future of biotechnology promises unprecedented breakthroughs that may redefine health, sustainability, and human evolution.

+123-456-7890

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Polímeros Naturais

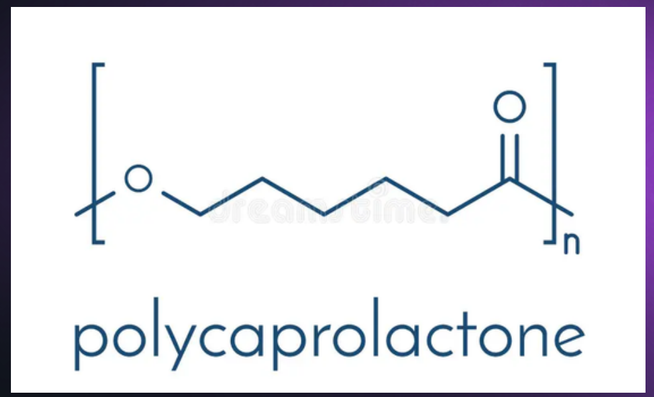
Mimetizam a matriz extracelular (MEC) nativa, garantindo alta bioatividade celular.

X

Polímeros Sintéticos

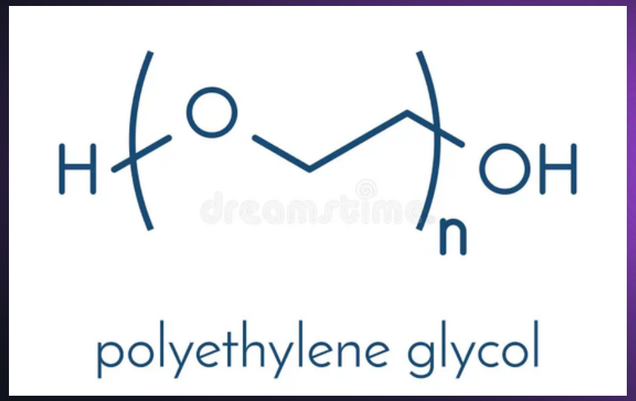
Degradação previsível e excelentes propriedades mecânicas para tecidos rígidos.

Policaprolactona (PCL)



Semicristalino e hidrofóbico. Ideal para tecidos rígidos (ósseos) por sua alta resistência mecânica e baixa toxicidade.

Polietilenoglicol (PEG)



Altamente hidrofílico. Melhora a retenção de água e o microambiente celular, permitindo reticulação por luz.

Polímeros Naturais

Mimetizam a matriz extracelular (MEC) nativa, garantindo alta bioatividade celular.

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Polímeros Sintéticos

Degradação previsível e excelentes propriedades mecânicas para tecidos rígidos.

Hidrogéis e Matriz Biomimética

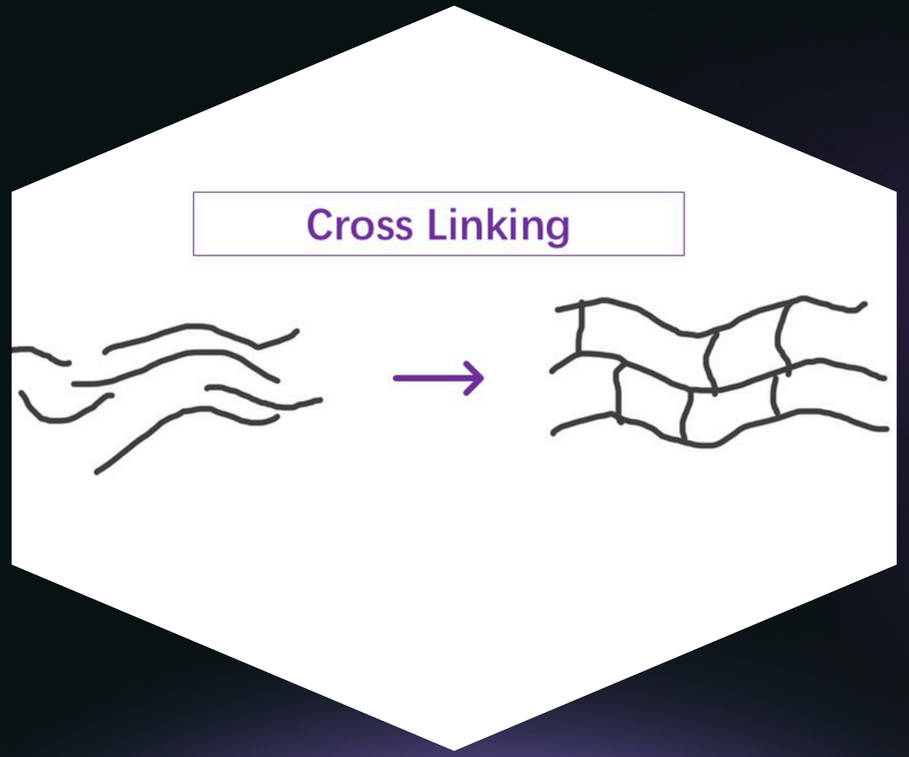
Redes 3D Hidrofílicas

Capazes de reter grandes volumes de água, simulando o microambiente biológico natural.

Difusão Contínua de Massa

Essencial para garantir o transporte adequado de oxigênio e nutrientes vitais às células.

Cinética de Reticulação (Crosslinking)



O QUE É O CROSSLINKING?

Criação de ligações químicas ou físicas entre as cadeias poliméricas para fixar a forma do órgão impresso.

VIAS DE RETICULAÇÃO

Processos físicos (interações térmicas ou iônicas) ou químicos (como a fotopolimerização).

DESAFIO NA ENG. QUÍMICA

Otimizar a taxa de reação para equilibrar a rigidez mecânica com a viabilidade celular sob cisalhamento.